



GLA
UNIVERSITY
MATHURA
Recognised by UGC Under Section 2(f)

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12-B Status from UGC

DEPARTMENT OF COMPUTER ENGINEERING & APPLICATIONS
Institute of Engineering & Technology

Lab Manual

**Subject Name & Code: OPERATING SYSTEM LAB
(BCAC 0818)**

Course: BCA

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Year: II

Semester: III

Session: 2024-25

OPERATING SYSTEM LAB (BCAC 0818)

Prerequisite: Familiarity with programming constructs

Credits: 02

Semester III

L-T-P : 0-0-2

Prac. No.	Program	Page No.
1	Case Study On Open Standard and Software: Introduction to Linux operating system and Preparing for Installation – Installation Checklist, Hardware Requirements, Partitioning, Installation problems, Working with the System, Shells and Utilities, Linux commands, File Handling using vi editor, Getting familiar with shell scripts.	4
2	Implement the following basic commands (with options) used in UNIX/LINUX OS: a) cp b) mv c) sort d) cut e) who f) whoami g) ps h) kill i) bc j) top k) grep l) chmod	8
3	Implement the following basic commands (with options) used in UNIX/LINUX OS: a) ls b) mkdir c) cd d) cat e) man f) date g) cal h) rm i) rmdir j) head k) tail l) pwd	13
4	Write a shell script to find whether a number is even or odd.	20
5	Write a shell script to check if the number entered at the command line is prime or not.	21
6	Write a shell script to input the name of a file as command line argument and display whether it is a file, a directory or anything else.	22
7	Write a menu driven shell script, which will print the following menu and execute the given task. • Display a calendar of current month • Display today's date and time. • Display username those are currently logged in the system. • Display your name at the given x,y position. • Display your terminal number.	23
8	Write a shell script to accept a login name. If not, a valid login name display message – “Entered login name is invalid”.	25
9	Write a shell script to compare two files and if found equal asks the user to delete the duplicate file.	26
10	Write a shell script to check whether the file have all the permissions or not.	28
11	Write a shell script to input the marks of a student in three subjects and determine their grade.	29
12	Write a shell script to display the Fibonacci series up to a specified limit n.	31
13	Write a shell script to display the multiplication table for a given number.	32
14	Write a shell script to reverse the number entered by the user.	33

15	Write a shell script to determine whether a given number is an Armstrong number.	34
16	Write a shell script to print a half-pyramid pattern using numbers.	36
17	Simulate FCFS CPU scheduling algorithm using shell script.	37
18	Simulate SJF CPU scheduling algorithm using shell script.	39
19	Simulate Priority CPU scheduling algorithm using shell script.	43
20	Simulate Round Robin CPU scheduling algorithm using shell script.	46
21	Simulate FIFO page replacement algorithm using shell script.	48
22	Simulate LRU page replacement algorithm using shell script.	51

****THE END****